Lesson notes · for you

Build a circuit: unit check

IoT · Level 2 · Unit check

Time
45 min

Materials
Paper and a pen. No device needed.

Learning goals

- Make it light up
- Switches that decide
- Talk in binary
- Meet the microcontroller

Activity: Quiz yourself

1. Answer each question on paper first, without looking back
2. Check your answers and read every explanation
3. Redo the lab from the lesson you found hardest
4. Explain one big idea to a friend or someone at home

Discussion

- Which idea surprised you most, and why?
- Where will you notice this at home or in town this week?
- What would you like to build or try next?

Check yourself

- One thing I can now explain, and one thing I want to try

Try it this week
Choose one idea from this course and show it to someone at home: build it, draw it or explain it in two minutes.

Worksheet

Build a circuit: unit check

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Name _____ Class _____ Date _____

1 **Grace's LED stays dark. The battery is new, the resistor is 100 Ω and the loop is closed. What should she check first?**

◯ Whether to take out the resistor ◯ That the LED's long leg goes to +

◯ Whether a bigger battery would help

2 **Match each problem to its fix.**

An LED burnt out in a flash	• • • •	Strip the wire end
The fan runs in empty rooms	• • • •	Put the program in a forever loop
The light sensor misses small changes	• • • •	Read it with more bits
The night light never comes on at night	• • • •	Change OR to AND
The bulb stays dark: a wire end is still in plastic	• • • •	Add a resistor

3 **A red LED on a 4.5 V battery. Work out the resistor.**

4.5 V minus 2 V for the LED leaves _____ V. For 10 mA, the resistor should be about _____ Ω.

Word bank: 4.5 · 250 · 450 · 2.5

4 **A rule with three inputs has 8 rows in its truth table.**

◯ True ◯ False

5 **The school bell should ring at 10:30, but only on school days. Which rule?**

◯ 10:30 OR school day ◯ 10:30 AND NOT school day ◯ 10:30 AND school day